Deep Learning | Action Recognition | Sports Analytics

Fine-Grained Sport Action Recognition

with Twin Spatio-Temporal CNNs

Application to Table Tennis Stroke Classification

Martin, Benois-Pineau, Péteri & Morlier

Multimedia Tools and Applications (2020)

91.4%	20	Twin
ACCURACY	CLASSES	NETWORK

FUNDAMENTALS

What is a CNN?

Convolutional Neural Networks are deep learning architectures inspired by the visual cortex. They automatically learn hierarchical features from raw pixels—from edges to complex patterns—**without manual feature engineering**.

How They Work

1. **Convolution:** Filters detect patterns
2. **Activation:** Non-linearity enables learning
3. **Pooling:** Reduces dimensionality
4. **Classification:** Makes predictions

Applications

- Image Classification
- Face Recognition
- Medical Imaging
- Object Detection
- Video Analysis
- Sports Analytics

Key advantage: CNNs leverage spatial structure through shared weights and local connectivity, making them vastly more efficient than fully-connected networks for visual tasks.

ARCHITECTURE

How CNNs Work

Convolution
1

Filters detect local patterns

```
Output = Input * Kernel
```

ReLU
2

Non-linearity enables complex learning

```
f(x) = max(0, x)
```

Pooling
3

Reduces dimensions

```
Max/Avg(region)
```

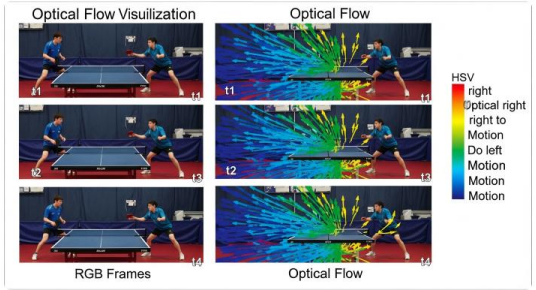
FC Layer
4

Final classification

```
Softmax(Wx + b)
```

3D Convolutions for Video

Standard 2D CNNs process spatial info. **3D convolutions** use kernels (W×H×T) to capture both *what appears* and *how it moves*across frames—essential for action recognition.



IMPORTANT

Ethical Issues

Data Bias

Unrepresentative datasets perpetuate societal biases

Privacy

Video systems collect sensitive personal information

Accountability

Who is responsible when AI systems make mistakes?

Potential Risks

- Discriminatory outcomes for minorities
- Mass surveillance without consent
- Black-box decisions affecting people

Mitigation Strategies

- ✓ Use diverse, balanced training data
- ✓ Implement transparency and explainability
- ✓ Maintain human oversight in decisions

RESEARCH

Goals & Motivation

Primary Goal

Classify **20 different table tennis strokes** from video footage to enable automated performance analysis and coaching feedback.

The Challenge

Table tennis strokes have **low inter-class variability** – different strokes look visually similar, making classification harder than typical tasks.

Markerless
Natural conditions

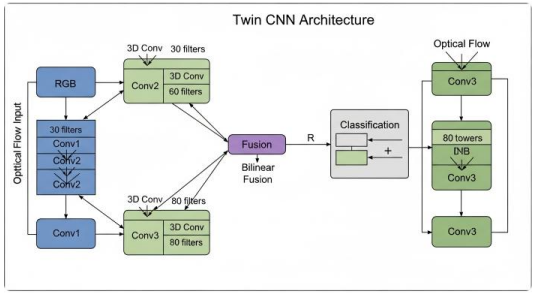
Real-World Data
TTStroke-21

Improve Training
Coach feedback

Why CNN is suitable: 3D CNNs capture spatial features (player pose, racket position) plus temporal features (motion dynamics). Twin architecture processes RGB and Optical Flow for comprehensive feature extraction.

ARCHITECTURE

Twin Spatio-Temporal CNN



Twin Streams

Parallel processing of RGB frames and Optical Flow for motion + appearance

3D Convolutions

3×3×3 kernels capture spatio-temporal patterns across 100 frames

Middle Fusion

Bilinear interpolation combines streams before classification

Input Size	Conv Filters	FC Layer	Output
120×120×100	30→60→80	500 units	21 classes

DATASET

TTStroke-21 Dataset

129	675K	120	17
Videos	Frames	FPS	Players

Stroke Categories (20 Total)

DEFENSIVE (6)

- BH/FH Backspin
- BH/FH Block
- BH/FH Push

OFFENSIVE (6)

- BH/FH Flip
- BH/FH Hit
- BH/FH Loop

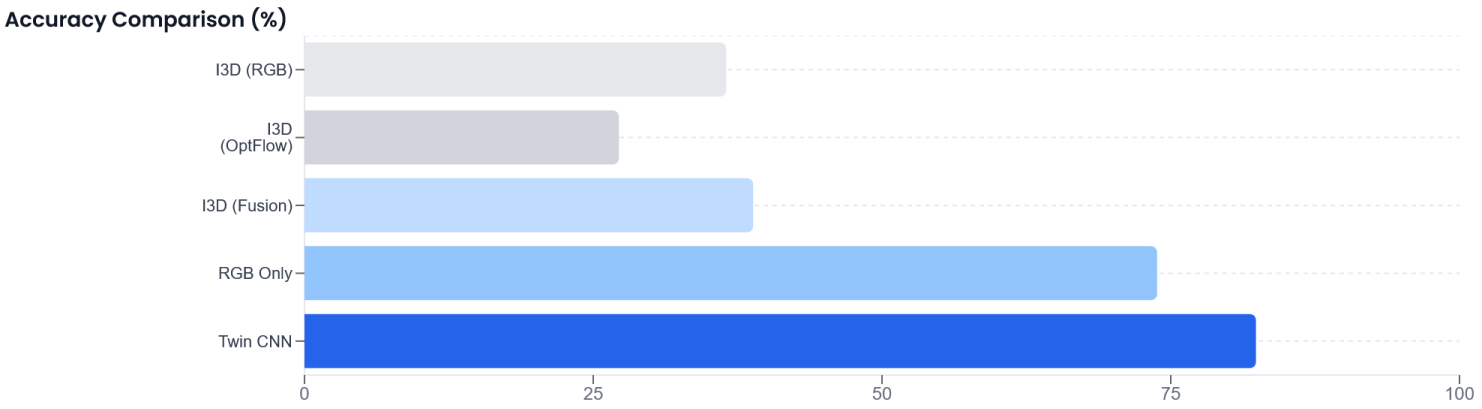
SERVE (8)

- BH/FH Backspin
- BH/FH Topspin
- BH/FH Sidespin
- BH/FH Loop

Details: University of Bordeaux recordings in markerless, natural conditions. Expert annotations. Avg stroke duration 1.47s (range: 0.83s–2.31s)

RESULTS

Classification Performance



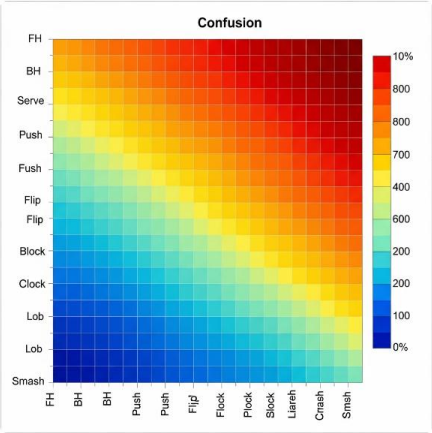
BEST RESULT	IMPROVEMENT	VS I3D
91.4%	**+48.3%**	**2.1×**

Key Findings

- Intermediate fusion outperforms late fusion approach
- Shallow networks better for specialized, limited data
- RGB stream more discriminative than optical flow alone

RESULTS

Confusion Matrix



Reading the Matrix

- **Diagonal (warm):** Correct classifications
- **Off-diagonal (cool):** Misclassifications

FH/BH: Forehand / Backhand strokes

D/O/S: Defensive / Offensive / Serve

Key Observations

- Strong diagonal = 91.4% overall accuracy
- Most errors between similar stroke types
- FH/BH distinction well-learned
- Serves classified with high precision

REFLECTION

My CNN Lab Experience

What I Did

Worked with CNNs for image classification using PyTorch/TensorFlow frameworks.

Included preprocessing data, defining network layers, training models, and evaluating with confusion matrices and metrics.

Key Learnings

- Features learned automatically via backpropagation
- Data augmentation critical for generalization
- Hyperparameter tuning significantly impacts results
- Validation monitoring prevents overfitting

Connection to This Paper

This paper helped me appreciate real-world CNN challenges. The researchers faced issues I encountered—like overfitting on specialized data and careful preprocessing needs. Their 3D convolutions and Twin architecture showed me how basic concepts extend to complex video tasks in practical applications.

CITATIONS

References

[1] Fine grained sport action recognition with Twin spatio-temporal CNNs
Martin, P.E., et al.
Multimedia Tools and Applications, 2020

Primary

[2] Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset
Carreira, J., & Zisserman, A.
CVPR 2017

[3] Two-Stream Convolutional Networks for Action Recognition
Simonyan, K., & Zisserman, A.
NIPS 2014

[4] UCF101: A Dataset of 101 Human Actions Classes
Soomro, K., et al.
arXiv:1212.0402, 2012

[5] Long-term Temporal Convolutions for Action Recognition
Varol, G., Laptev, I., & Schmid, C.
IEEE TPAMI, 2018

SUMMARY

Key Takeaways

91.4%
Twin CNN Accuracy
Outperforms deep baselines by 48.3%

3D Conv
Spatio-Temporal
Captures motion and appearance

Ethics
Critical for AI
Bias, privacy, accountability

Key Insights

- Automatic feature learning eliminates manual engineering
- Shallow networks can outperform deep ones on limited data
- Architecture design impacts specialized domain performance
- Multi-stream fusion captures complementary information

Thank You

Questions?